

KSRCE INTERNSHIP COHORT - 15-DAY PROGRAM

AI-Powered Full-Stack Product Scaffolding

A 15-Day for AI-Assisted Full-Stack Engineering

15-Day Daywise Schedule

Day & Duration	Curated Topic & Sub-topics	First Principles & Value Add
PART I: COHORT ONBOARDING & SETUP (DAYS 1-2)		
Day 1 (2.0 Hours)	Program Intro & LaaS Portal Guide - Cohort introductions & team formations - Signing up on the LaaS portal - Launching remote desktops & booking slots - ZFS home folder limits & hardware checks	Establishes immediate compute access. Interns learn how to launch and navigate isolated remote desktop sessions on local RTX 5090 machines.
Day 2 (2.0 Hours)	Vibe-Coding & Antigravity Setup - Understanding the 'Vibe-Coding' paradigm shift - Downloading & installing the Antigravity developer environment - Workspace configurations & environment variables - Running basic agent commands & background checks	Prepares the local development environment, transitioning students from manual syntax writing to AI-driven command execution.
PART II: PRODUCT WORKFLOWS & CONTEXT ENGINEERING (DAYS 3-6)		

Day 3 (1.5 Hours) [1h Teach, 0.5h Doubt]	Product Dev Workflow & Harness (Part 1) ▪ AI-assisted rapid prototyping models ▪ Setting up core repository directories ▪ Creating temporary code scratchpads ▪ Tracking file edits & compilation logs	Establishes the write-run-fix loop, showing how to coordinate AI outputs with real-time execution outputs to audit code changes.
Day 4 (1.5 Hours) [1h Teach, 0.5h Doubt]	Product Dev Workflow & Harness (Part 2) ▪ Harness pipeline scripting & configs ▪ Running automated code verification steps ▪ Refactoring code blocks via agent commands ▪ Managing prompt-compile cycles	Enables test-driven agent logic, showing interns how to instruct agents to build, run, and self-correct files based on terminal execution errors.
Day 5 (1.5 Hours) [1h Teach, 0.5h Doubt]	Context Engineering: Rules & Skills (Part 1) ▪ System prompts vs. user input structures ▪ Designing custom instructions files ▪ Controlling token boundaries ▪ Context size optimization techniques	Reduces model hallucinations. Interns learn to restrict and focus model contexts so the AI pair programmer outputs accurate results.
Day 6 (1.5 Hours) [1h Teach, 0.5h Doubt]	Context Engineering: Rules & Skills (Part 2) ▪ Writing custom .qoder workspace rules ▪ Injecting team coding style guides ▪ Mapping specific workspace tools (skills) ▪ Enforcing API/schema coding boundaries	Ensures generated code aligns with project structures, database requirements, and repository parameters automatically.

PART III: CORE WEB APIs, DATA & LLMs (DAYS 7-10)

Day 7 (1.5 Hours) [1h Teach, 0.5h Doubt]	Generative AI: LLM Fundamentals & APIs ▪ Attention & Transformer architecture basics ▪ Temperature parameters & context constraints ▪ Securing API tokens & handling connection retries ▪ Enforcing outputs via JSON mode	Transitions students from standard chat apps to calling LLM APIs programmatically within Python or Node.js backend handlers.
---	---	--

Day 8 (1.5 Hours) [1h Teach, 0.5h Doubt]	Vector Databases, RAG & Prompting ▪ Text embeddings & mathematical token distance ▪ Chroma DB & pgvector similarity search ▪ Implementing text chunking & overlap logic ▪ RAG query context injection pipelines	Enables semantic searches. Interns search local manuals, databases, and code directories semantically without expensive model fine-tuning costs.
Day 9 (1.5 Hours) [1h Teach, 0.5h Doubt]	Agentic Systems: Harness, Rules & Skills ▪ Defining Agent autonomy: Reason-Act (ReAct) loop ▪ Function schema declaration to LLMs ▪ Managing states across recursive agent loops ▪ Setting sandbox tool execution parameters	Builds autonomous agents that can evaluate outputs, call external system tools, and iterate to solve complex multi-step tasks.
Day 10 (1.5 Hours) [1h Teach, 0.5h Doubt]	Multi-Agent Workflows & Orchestrations ▪ Visual workflow automation using n8n ▪ Coordinating event-based triggers (webhooks, schedules) ▪ Chaining multi-agent inputs/outputs ▪ Orchestrating agent hierarchies	Teaches enterprise workflow integrations, showing how to connect databases, REST APIs, and agent decisions into clean automated pipelines.
PART IV: OPEN PROJECT COACHING STAGE (DAYS 11-15)		
Days 11–15 (Open stage daily)	Ongoing Project Mentoring & Doubt Clearing ▪ Daily team progress & block checks ▪ Workspace debugging & environment fixes ▪ API routing & database connector audits ▪ Live code-reviews & optimization feedback ▪ Final project demo pitching rehearsals	Consolidates all program knowledge, providing direct engineering support as teams build their POC asynchronously, culminating in the Day 15 showcase.

Resource Allocation & Data Ingestion Scheme

Each team is allocated a strict compute layout. One team member receives a Blaze instance (4GB VRAM) to host the core integrations, while the remaining members receive Spark instances (2GB VRAM) to develop their specific modules. Since containers run on central GPU hosts, there is no support for live webcam/mic feeds. All media feeds are simulated via pre-recorded files or public URLs:

Ingestion Method	Operational Workflow
File Upload	Intern records media on smartphone -> uploads via LaaS file transfer tool -> stored in /home/ubuntu ZFS volume.
URL Ingestion	Containers download public links using yt-dlp, wget, or index RTSP streams from public URL inputs.
Text Input	Students write structured descriptions or prompt text directly into apps running inside the Selkies desktop.

Collaborative Projects: Specifications for 14 Teams

1. PixelRevive — AI Photo Restoration Studio

Difficulty: Standard

- **One-liner:** Upload a damaged, old, black-and-white family photo -> get back a restored, colorized, 4K version.
- **Demo Moment:** Show a torn 1970s family photo -> watch AI fix scratches, add color, upscale to HD. Before/after reveal.
- **Data Input:** Student uploads photo via LaaS web UI

Module	Member	GPU Model (VRAM)	What They Build / Responsibility
Scratch Removal	Member 1	U-Net-small (~800 MB)	Inpainting model to fill scratches, tears, water damage
Colorization	Member 2	DeOldify (~100 MB)	Auto-colorize B&W photos with plausible colors
Super-Resolution	Member 3	Real-ESRGAN x4 (~500 MB)	Upscale restored photo to 4K quality
Web Studio + AI Agent	Member 4	ChromaDB + Embeddings (~300 MB)	Upload UI, side-by-side comparison, batch processing, AI describes changes

2. SnapChef — Fridge-to-Recipe AI

Difficulty: Standard

- **One-liner:** Snap a photo of your fridge -> AI identifies every ingredient -> generates a personalized recipe.
- **Demo Moment:** Open a fridge, take a photo, upload it -> watch ingredients get detected with bounding boxes -> full recipe appears.
- **Data Input:** Student uploads fridge/pantry photo

Module	Member	GPU Model (VRAM)	What They Build / Responsibility
Ingredient Detection	Member 1	YOLOv8 Small (~1.5 GB)	Fine-tune to detect 50+ common food items
Recipe Generation	Member 2	Qwen2-1.5B-Q4 on Blaze (~1.1 GB)	Generate recipes from detected ingredients
Nutrition + Preferences	Member 3	ChromaDB + Embeddings (~300 MB)	Dietary filters, allergy checks, nutrition scoring
Cook-Along UI	Member 4	Frontend UI (No GPU model)	Step-by-step interface, timer, ingredient checklist

3. SoundForge — AI Music & Beat Generator

Difficulty: Standard

- **One-liner:** Describe a mood in text -> AI generates original music. Layer tracks, add beats, export.
- **Demo Moment:** Type 'chill lo-fi with Tamil folk influence' -> music starts playing. Layer a beat. Export the mix.
- **Data Input:** Text description (typed in desktop)

Module	Member	GPU Model (VRAM)	What They Build / Responsibility
Music Generation	Member 1	MusicGen Small on Blaze (~2.5 GB)	Text-to-music, genre/mood conditioning
Beat Detection & Mixing	Member 2	Audio DSP + PyTorch (~500 MB)	BPM detection, beat alignment, audio mixing
Audio Effects	Member 3	Neural audio effects (~400 MB)	Reverb, EQ, style transfer on audio
Studio UI	Member 4	Frontend (Web Audio API)	Multi-track editor, waveform viz, export to MP3

4. FormCheck — AI Exercise Form Analyzer

Difficulty: Standard

- **One-liner:** Upload a video of yourself exercising -> AI tracks your pose -> generates a form report with corrections.
- **Demo Moment:** Upload a squat video -> watch skeleton overlay -> 'knee angle: 72 (should be 90)' at timestamps

-> AI tips.

- **Data Input:** Student uploads exercise video (recorded on phone)

Module	Member	GPU Model (VRAM)	What They Build / Responsibility
Video Processing	Member 1	OpenCV frame extraction	Extract frames, manage video pipeline
Pose Estimation	Member 2	MediaPipe Pose (~300 MB)	Real-time 33-point body tracking on GPU
Form Analysis	Member 3	Custom angle classifier (~200 MB)	Joint angles, correct-form comparison, rep counting
Report UI	Member 4	Frontend Canvas overlay	Video playback + skeleton overlay, angle graphs, AI tips

5. HawkEye — AI Video Intelligence Engine

Difficulty: Standard

- **One-liner:** Paste a YouTube URL or upload video -> AI detects objects, counts people, finds anomalies -> searchable event timeline.
- **Demo Moment:** Paste a traffic camera YouTube URL -> container pulls stream -> real-time detection overlay -> searchable timeline.
- **Data Input:** Paste URL (yt-dlp) or upload .mp4

Module	Member	GPU Model (VRAM)	What They Build / Responsibility
Video Ingestion	Member 1	yt-dlp + OpenCV	URL download, frame extraction, video management
Frame Detection	Member 2	YOLOv8 Nano (~1 GB)	Per-frame object/person/vehicle detection on GPU
Event Intelligence	Member 3	ChromaDB + Embeddings (~300 MB)	Anomaly detection, object tracking, event classification
Timeline Dashboard	Member 4	Frontend Timeline	Searchable event log, video scrubber with annotations, export

6. GameBrain — AI That Learns to Play Classic Games

Difficulty: Challenging

- **One-liner:** Pick a game (Snake, Pong, Flappy Bird) -> watch AI train from zero to superhuman in real-time.
- **Demo Moment:** Start training on Snake -> AI crashes -> 2 minutes later it plays perfectly. AI vs. Human challenge mode.
- **Data Input:** Internal (no external input - game runs inside the container)

Module	Member	GPU Model (VRAM)	What They Build / Responsibility
--------	--------	------------------	----------------------------------

RL Training Engine	Member 1	DQN/PPO on GPU (~800 MB)	Train agents with Deep Q-Learning on GPU
Game Environments	Member 2	PyTorch + Gymnasium (~500 MB)	GPU-accelerated Snake, Pong, Flappy Bird
Visualization	Member 3	State analysis embeddings (~300 MB)	Neural net viz, learning curves, state-action heatmaps
Arena UI	Member 4	Frontend Canvas/WebGL	Game viewer, training controls, leaderboard, AI vs. Human

7. ArtForge — AI Art & Style Studio

Difficulty: Standard

- **One-liner:** Upload any photo -> apply 10+ artistic styles (Van Gogh, Picasso, anime) -> adjust intensity -> export.
- **Demo Moment:** Upload a selfie -> click through 10 styles -> renders in <2 seconds -> drag slider -> export.
- **Data Input:** Student uploads photo

Module	Member	GPU Model (VRAM)	What They Build / Responsibility
Style Transfer Engine	Member 1	Fast Neural Style (~1 GB)	10+ pre-trained style models on GPU
Batch Processing	Member 2	GPU pipeline (~500 MB)	Multi-style simultaneously, adjustable intensity
AI Art Curation	Member 3	ChromaDB + Embeddings (~300 MB)	'Find similar styles', gallery, style recommendation
Studio UI	Member 4	Frontend Gallery	Drag-drop upload, style gallery, intensity slider, comparison

8. EchoScribe — AI Video Transcription & Study Platform

Difficulty: Standard

- **One-liner:** Upload a lecture video or paste YouTube URL -> AI transcribes, generates chapters, flashcards, and quizzes.
- **Demo Moment:** Paste lecture URL -> real-time transcription -> chapters auto-generate -> click for AI flashcards.
- **Data Input:** Upload video or paste YouTube URL (yt-dlp)

Module	Member	GPU Model (VRAM)	What They Build / Responsibility
Transcription	Member 1	Whisper Small (~2 GB)	Speech-to-text with timestamps + speaker diarization
Intelligence Layer	Member 2	ChromaDB + Embeddings (~300 MB)	Topic segmentation, chapter detection, key moments

Study Tools	Member 3	Qwen2-1.5B-Q4 on Blaze (~1.1 GB)	Flashcards, summaries, quiz questions from transcript
Video Player UI	Member 4	Frontend Video Player	Synced video + transcript, chapter sidebar, flashcard viewer

9. VoxLingua — Voice Translator

Difficulty: Standard

- **One-liner:** Upload audio in Tamil -> AI transcribes, translates to English, and speaks it back.
- **Demo Moment:** Upload a Tamil voice recording -> see transcription appear -> see English translation -> hear it spoken aloud.
- **Data Input:** Student uploads audio file (.wav/.mp3, recorded on phone)

Module	Member	GPU Model (VRAM)	What They Build / Responsibility
Speech-to-Text	Member 1	Whisper Small (~2 GB)	Tamil + English + Hindi transcription on GPU
Translation Engine	Member 2	Qwen2-1.5B-Q4 on Blaze (~1.1 GB)	Context-aware translation
Text-to-Speech	Member 3	Piper TTS (GPU, ~500 MB)	Natural voice output in target language
Conversation UI	Member 4	Frontend Audio Player	Audio player, transcript view, translation panel, playback

10. MemeForge — Context-Aware AI Meme Generator

Difficulty: Standard

- **One-liner:** Upload any image -> AI understands the context -> generates genuinely funny captions.
- **Demo Moment:** Upload a photo of the professor -> AI generates a meme about assignments. The room erupts.
- **Data Input:** Student uploads image or picks from trending templates

Module	Member	GPU Model (VRAM)	What They Build / Responsibility
Image Understanding	Member 1	CLIP ViT-B/32 (~400 MB) + YOLOv8 Nano (~1 GB)	Detect objects, scene context in image
Caption Generation	Member 2	Qwen2-1.5B-Q4 on Blaze (~1.1 GB)	Contextually relevant, funny captions
Template Engine	Member 3	ChromaDB + Embeddings (~300 MB)	Match image to meme templates, text placement
Meme Studio UI	Member 4	Frontend Canvas Workspace	Upload/template picker, caption editor, text drag-

			drop
--	--	--	------

11. Scene3D — Single-Image 3D Depth Scanner

Difficulty: Challenging

- **One-liner:** Upload any photo -> AI estimates depth for every pixel -> generates a 3D point cloud you can rotate and fly through.
- **Demo Moment:** Upload a landscape photo -> watch it transform into a rotating 3D scene. Fly through the depth.
- **Data Input:** Student uploads photo

Module	Member	GPU Model (VRAM)	What They Build / Responsibility
Depth Estimation	Member 1	Depth Anything V2 Small (~500 MB)	Per-pixel depth prediction from single image
3D Point Cloud	Member 2	PyTorch3D + GPU (~500 MB)	Depth map -> 3D point cloud, mesh generation
Scene Enhancement	Member 3	Real-ESRGAN (~500 MB)	Upscale input for better depth, semantic understanding
3D Viewer UI	Member 4	Frontend (Three.js)	Interactive 3D viewer, orbit/fly controls, depth slider

12. StoryQuest — AI Interactive Adventure Game

Difficulty: Challenging

- **One-liner:** An AI Dungeon Master that creates interactive adventures with voice narration AND generates an image for every scene.
- **Demo Moment:** 'You are in a dark forest. Which way?' -> choose left -> AI narrates -> generates scene image.
- **Data Input:** Text choices (typed in desktop)

Module	Member	GPU Model (VRAM)	What They Build / Responsibility
Story Engine	Member 1	Qwen2-1.5B-Q4 on Blaze (~1.1 GB)	Dynamic story, branching narratives, character consistency
Scene Illustration	Member 2	SD 1.5 optimized on Blaze (~3.5 GB)	Generate image for each scene description
Voice Narration	Member 3	Whisper Base (~1 GB) + Piper TTS	Voice input for choices + AI narration
Game UI	Member 4	Frontend Gaming Board	Text adventure, scene illustrations, choice buttons, story map

13. SignReader — Video Sign Language Translator

Difficulty: Challenging

- **One-liner:** Upload a video of someone signing -> AI tracks hand poses -> translates gestures to text -> speaks the translation.
- **Demo Moment:** Upload a 30-second signing video -> watch hand tracking overlay -> see words appear -> hear spoken translation.
- **Data Input:** Student uploads signing video (recorded on phone)

Module	Member	GPU Model (VRAM)	What They Build / Responsibility
Video Processing	Member 1	OpenCV Frame Ingestion	Extract frames, manage video pipeline
Hand Tracking	Member 2	MediaPipe Hands (~300 MB)	Real-time 21-point hand landmark detection on GPU
Gesture Recognition	Member 3	LSTM on pose sequences (~500 MB)	Classify gesture sequences into words/phrases
Translation UI	Member 4	Frontend Subtitle Player	Video player + synced translation, vocabulary lookup, TTS

14. PixelPlayground — Creative AI Toolkit (5-person team)

Difficulty: Standard+

- **One-liner:** A Swiss Army knife of AI visual tools: background remover, style transfer, upscaler, colorizer, depth mapper — all in one app.
- **Demo Moment:** Upload photo -> remove background -> apply style -> upscale -> see depth map. Each tool = different GPU model.
- **Data Input:** Student uploads photos

Module	Member	GPU Model (VRAM)	What They Build / Responsibility
Background Removal	Member 1	TinySAM (~500 MB) + rembg	One-click bg removal, object cutout, bg swap
Style Transfer + Filters	Member 2	Fast Neural Style (~1 GB)	10+ art styles, adjustable intensity, real-time preview
Super-Res + Colorization	Member 3	Real-ESRGAN (~500 MB) + DeOldify (~100 MB)	4x upscaling, B&W colorization, photo enhancement
Depth + 3D Preview	Member 4	Depth Anything V2 (~500 MB)	Depth map, 3D parallax effect, depth-based blur
Unified UI + AI Agent	Member 5	ChromaDB + Embeddings (~300 MB)	Drag-drop canvas, tool palette, batch processing, AI suggests tools